

Data Analysis

Portfolio Project Challenge

Lung Cancer Prediction

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<PostgreSQL> <Excel> <PowerBI>

Business Problem/Goal of the Analysis

The goal of the analysis is to utilize the Lung Cancer Prediction Dataset to predict the likelihood of lung cancer in individuals based on demographic, medical, lifestyle, genetic, and environmental factors. The analysis aims to assist in identifying high-risk individuals, providing early detection insights, and informing personalized treatment strategies for lung cancer. The primary objective is to develop a predictive model that can accurately identify potential lung cancer patients, assess survival rates, and support healthcare providers in making data-driven decisions for patient care and treatment planning.

Objectives Of the Project:

- 1 **Predict Lung Cancer Diagnosis:** Build a model to predict lung cancer diagnosis.
- 2 **Feature Importance Identification:** Identify key factors influencing lung cancer diagnosis and prognosis.
- 3 **Survival Analysis:** Assess risk factors affecting lung cancer survival rates.
- 4 **Early Detection Strategy:** Develop strategies for early detection of lung cancer.
- 5 **Mortality Risk Prediction:** Predict mortality rates for lung cancer patients.
- 6 **Treatment Recommendations:** Recommend treatments based on patient risk factors and cancer stage.

Data Description

- **There are 220632 Rows and 24 Columns.**
- **Columns :** 'ID', 'Country', 'Population_Size', 'Age', 'Gender', 'Smoker', 'Years_of_Smoking', 'Cigarettes_per_Day', 'Passive_Smoker', 'Family_History', 'Lung_Cancer_Diagnosis', 'Cancer_Stage', 'Survival_Years', 'Adenocarcinoma_Type', 'Air_Pollution_Exposure', 'Occupational_Exposure', 'Indoor_Pollution', 'Healthcare_Access', 'Early_Detection', 'Treatment_Type', 'Developed_or_Developing', 'Annual_Lung_Cancer_Deaths', 'Lung_Cancer_Prevalence_Rate', 'Mortality_Rate'
- **Data Types :** String → 15, Integer → 7 , Float → 2

Column	mean	sd	p0	p25	p50	p75	p100
ID	110300	63690	0	55160	110300	165500	220600
Population_Size	229.8	349.2	54	83	113	206	1400
Age	52.52	19.08	20	36	53	69	85
Years_of_Smoking	8.175	12.38	0	0	0	15	40
Cigarettes_per_Day	7.008	9.802	0	0	0	14	30
Survival_Years	0.2235	1.231	0	0	0	0	10
Annual_Lung_Cancer_Deaths	63930	130700	10000	23000	30000	45000	690000
Lung_Cancer_Prevalence_Rate	1.502	0.578	0.5	1	1.5	2	2.5
Mortality_Rate	3.05	14.92	0	0	0	0	90

Data Preprocessing:

- **Removing Duplicates:**
 - Duplicate records are identified and removed to ensure data consistency.
- **Handling Missing Values:**
 - Numeric missing values are filled with the median of their respective columns to prevent skewness in the dataset.
 - Categorical missing values are filled with the mode (most frequent value) to maintain data consistency.
- **Ensuring Consistent Data Types and Cleaning Text Columns:**
 - If the 'Gender' column exists, string formatting is applied to standardize values by stripping extra spaces and capitalizing the first letter.
- **Mapping Boolean-Like Fields:**
 - Columns such as 'Smoker', 'Passive_Smoker', and 'Early_Detection' contain Yes/No values.
 - These values are mapped to numeric representations (Yes → 1, No → 0) to facilitate further analysis.
 - Any missing values in these columns are filled with 0 as a default value.

<PostgreSQL>

Basic Level

1.Retrieve all records for individuals diagnosed with lung cancer:

SELECT *FROM lung_cancer_data WHERE Lung_Cancer_Diagnosis = 'Yes';

	id [PK] integer	country character varying (255)	population_size integer	age integer	gender character varying (10)	smoker character varying (3)
1	26	Pakistan	225	40	Female	1
2	32	Nigeria	206	55	Male	1
3	33	Turkey	85	33	Male	1
4	93	UK	67	61	Male	1

4.Retrieve the average number of cigarettes smoked per day by smokers:

SELECT AVG(Cigarettes_per_Day) AS avg_cigarettes FROM lung_cancer_data WHERE Smoker = ‘1’;

	avg_cigarettes numeric
1	17.5012961139221879

7.Count the number of people diagnosed with lung cancer by gender:

SELECT Gender, COUNT(*) AS count FROM lung_cancer_data WHERE Lung_Cancer_Diagnosis = 'Yes'GROUP BY Gender;

	gender character varying (10)	count bigint
	Female	362
	Male	533

2.Count the number of smokers and non-smokers:

SELECT Smoker, COUNT(*) AS count FROM lung_cancer_data GROUP BY Smoker;

	smoker character varying (3)	count bigint
1	0	132291
2	1	88341

3.List all unique cancer stages present in the dataset:

SELECT DISTINCT Cancer_Stage FROM lung_cancer_data;

	cancer_stage character varying (50)
1	Stage 1
2	Stage 2
3	Stage 3
4	Stage 4

5.Count the number of people exposed to high air pollution:

SELECT COUNT(*) AS high_pollution_count FROM lung_cancer_data WHERE Air_Pollution_Exposure = 'High';

	high_pollution_count bigint
1	55108

8.Retrieve records of individuals older than 60 who are diagnosed with lung cancer:

SELECT *FROM lung_cancer_data WHERE Age > 60 AND Lung_Cancer_Diagnosis = 'Yes';

	id [PK] integer	country character varying (255)	population_size integer	age integer	gender character varying (10)	smoker character varying (3)
1	93	UK	67	61	Male	1
2	106	Ethiopia	120	70	Male	1
3	157	Germany	83	72	Male	1
4	188	Egypt	102	71	Male	1
5	227	Russia	145	85	Male	1
6	229	Turkey	85	66	Male	1
7	335	Indonesia	273	73	Male	1
8	848	Nigeria	206	72	Male	1
9	879	Thailand	70	81	Male	1

6.Find the top 5 countries with the highest lung cancer deaths:

SELECT Country, SUM(Annual_Lung_Cancer_Deaths) AS total_deaths FROM lung_cancer_data GROUP BY Country ORDER BY total_deaths DESC LIMIT 5;

	country character varying (255)	total_deaths bigint
1	China	6145830000
2	USA	1139970000
3	Japan	674175000
4	India	612640000
5	Russia	528240000

<PostgreSQL>

Intermediate Level

1.Find the percentage of smokers who developed lung cancer:

```
SELECT (COUNT(CASE WHEN Smoker = 1 AND Lung_Cancer_Diagnosis = 'Yes' THEN 1 END) * 100.0 /COUNT(CASE WHEN Smoker = 1 THEN 1 END)) AS smoker_cancer_percentage FROM lung_cancer_data;
```

	smoker_cancer_percentage numeric
1	7.0737256766393860

2.Calculate the average survival years based on cancer stages:

```
SELECT Cancer_Stage, AVG(Survival_Years) AS avg_survival_years FROM lung_cancer_data GROUP BY Cancer_Stage;
```

	cancer_stage character varying (50)	avg_survival_years numeric
1	Stage 1	5.4217252396166134
2	Stage 2	5.5969062784349409
3	Stage 3	5.5514874141876430
4	Stage 4	0.06075923347877678013

3.Count the number of lung cancer patients based on passive smoking:

```
SELECT Passive_Smoker, COUNT(*) AS count FROM lung_cancer_data WHERE Lung_Cancer_Diagnosis = 'Yes'GROUP BY Passive_Smoker;
```

	passive_smoker character varying (3)	count bigint
1	0	6226
2	1	2735

4.Find the country with the highest lung cancer prevalence rate:

```
SELECT Country FROM lung_cancer_data ORDER BY Lung_Cancer_Prevalence_Rate DESC LIMIT 1;
```

	country character varying (255)
1	Egypt

5.Identify the smoking years' impact on lung cancer:

```
SELECT Years_of_Smoking, COUNT(*) AS lung_cancer_count FROM lung_cancer_data WHERE Lung_Cancer_Diagnosis = 'Yes'GROUP BY Years_of_Smoking ORDER BY Years_of_Smoking;
```

	years_of_smoking integer	lung_cancer_count bigint
1	0	2712
2	1	142
3	2	163
4	3	164
5	4	143
6	5	154
7	6	158
8	7	147
9	8	141
10	9	179

6.Determine the mortality rate for patients with and without early detection:

```
SELECT Early_Detection, AVG(Mortality_Rate) AS avg_mortality_rate FROM lung_cancer_data GROUP BY Early_Detection;
```

	early_detection character varying (3)	avg_mortality_rate double precision
1	0	3.0374750335089007
2	1	3.0815400443947514

7.Group the lung cancer prevalence rate by developed vs. developing countries:

```
SELECT Developed_or_Developing, AVG(Lung_Cancer_Prevalence_Rate) AS avg_prevalence_rate FROM lung_cancer_data GROUP BY Developed_or_Developing;
```

	developed_or_developing character varying (50)	avg_prevalence_rate double precision
1	Developed	1.5017931217031197
2	Developing	1.502177404450925

<PostgreSQL>

Advanced Level

1. Identify the correlation between lung cancer prevalence and air pollution levels:

```
SELECT Air_Pollution_Exposure,
AVG(Lung_Cancer_Prevalence_Rate) AS
avg_prevalence_rate FROM lung_cancer_data GROUP BY
Air_Pollution_Exposure;
```

	air_pollution_exposure character varying (10)	avg_prevalence_rate double precision
1	High	1.503072875081652
2	Low	1.503043870897833
3	Medium	1.5011094165599956

4. Rank countries based on their mortality rate:

```
SELECT Country, AVG(Mortality_Rate) AS
avg_mortality_rate FROM lung_cancer_data GROUP BY
Country ORDER BY avg_mortality_rate DESC;
```

	country character varying (255)	avg_mortality_rate double precision
1	Ethiopia	3.4289047353760433
2	Japan	3.3237534764712424
3	Turkey	3.2601907204996654
4	USA	3.225872961569167
5	Nigeria	3.209238871899424
6	Myanmar	3.208582605737613

2. Find the average age of lung cancer patients for each country:

```
SELECT Country, AVG(Age) AS avg_age FROM
lung_cancer_data WHERE Lung_Cancer_Diagnosis =
'Yes' GROUP BY Country;
```

	country character varying (255)	avg_age numeric
1	Bangladesh	52.4321329639889197
2	Brazil	53.3767705382436261
3	China	52.8846153846153846
4	DR Congo	52.3516483516483516
5	Egypt	54.0404312668463612
6	Ethiopia	52.7726161369193154
7	France	52.5917808219178082

5. Determine if treatment type has a significant impact on survival years:

```
SELECT Treatment_Type, AVG(Survival_Years) AS
avg_survival_years FROM lung_cancer_data GROUP BY
Treatment_Type;
```

	treatment_type character varying (50)	avg_survival_years numeric
1	Chemotherapy	5.4192343604108310
2	Radiotherapy	5.4755555555555556
3	Surgery	0.11741120976692563818

8. Analyze the impact of early detection on survival years:

```
SELECT Early_Detection, AVG(Survival_Years) AS
avg_survival_years FROM lung_cancer_data GROUP BY
Early_Detection;
```

	early_detection character varying (3)	avg_survival_years numeric
1	0	0.22143562829976150472
2	1	0.22890843986454738411

3. Calculate the risk factor of lung cancer by smoker status, passive smoking, and family history:

```
SELECT Smoker, Passive_Smoker, Family_History, COUNT(*) AS
lung_cancer_count FROM lung_cancer_data WHERE
Lung_Cancer_Diagnosis = 'Yes' GROUP BY Smoker, Passive_Smoker,
Family_History;
```

	smoker character varying (3)	passive_smoker character varying (3)	family_history character varying (3)	lung_cancer_count bigint
1	0	0	No	1650
2	0	0	Yes	266
3	0	1	No	684
4	0	1	Yes	112
5	1	0	No	3677
6	1	0	Yes	633

6. Compare lung cancer prevalence in men vs. women across countries:

```
SELECT Country, Gender, AVG(Lung_Cancer_Prevalence_Rate)
AS avg_prevalence_rate FROM lung_cancer_data GROUP BY
Country, Gender;
```

	country character varying (255)	gender character varying (10)	avg_prevalence_rate double precision
1	Bangladesh	Female	1.5009943757030315
2	Bangladesh	Male	1.4997372029250426
3	Brazil	Female	1.4917988700564946
4	Brazil	Male	1.5237531428571476
5	China	Female	1.4942750000000002
6	China	Male	1.5050832039050386

7. Find how occupational exposure, smoking, and air pollution collectively impact lung cancer rates:

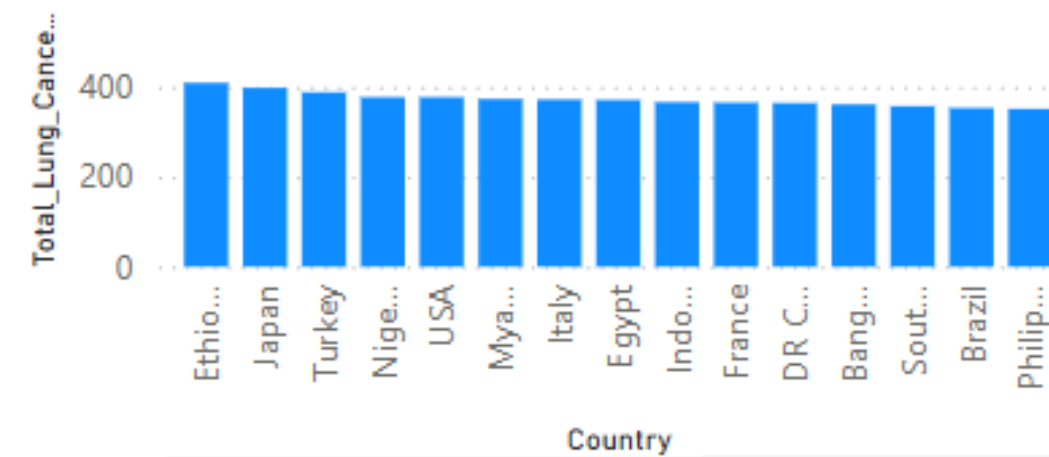
```
SELECT Occupational_Exposure, Smoker,
Air_Pollution_Exposure, COUNT(*) AS lung_cancer_count
FROM lung_cancer_data WHERE Lung_Cancer_Diagnosis =
'Yes' GROUP BY Occupational_Exposure, Smoker,
Air_Pollution_Exposure;
```

	occupational_exposure character varying (3)	smoker character varying (3)	air_pollution_exposure character varying (10)	lung_cancer_count bigint
1	No	0	High	511
2	No	0	Low	466
3	No	0	Medium	906
4	No	1	High	1024
5	No	1	Low	1077

Key Insights:-

- **Total Lung Cancer Cases:** 8,961 reported cases
- **Average Age of Lung Cancer Patients:** 52.66 years.
- **Percentage of Smokers with Lung Cancer:** 7% (0.07 in decimal form).
- **Mortality Rate:** 4.26M deaths associated with lung cancer
- **Males have a higher smoking impact score** than females, indicating a stronger correlation between smoking and lung cancer in men.
- The **mortality rate is influenced by air pollution levels**, with high pollution areas showing the most deaths.
- The **average age of lung cancer patients** is around 52.66 years, with a **peak in lung cancer cases between ages 45-60**.

Lung Cancer Cases by Country



Country

- ☐ Bangladesh
- ☐ Brazil

Gender

Female

Male

Age

20

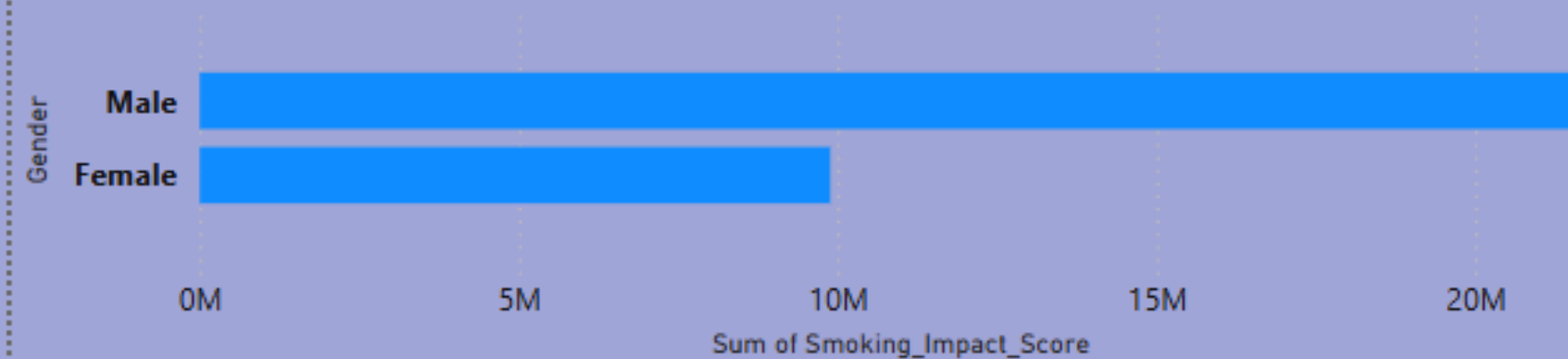
85

Smoker

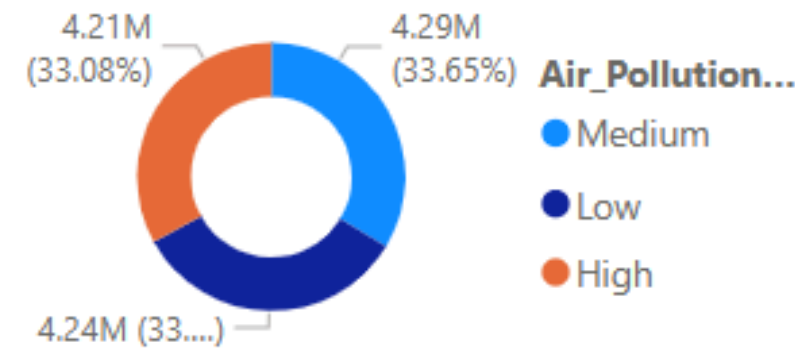
0

1

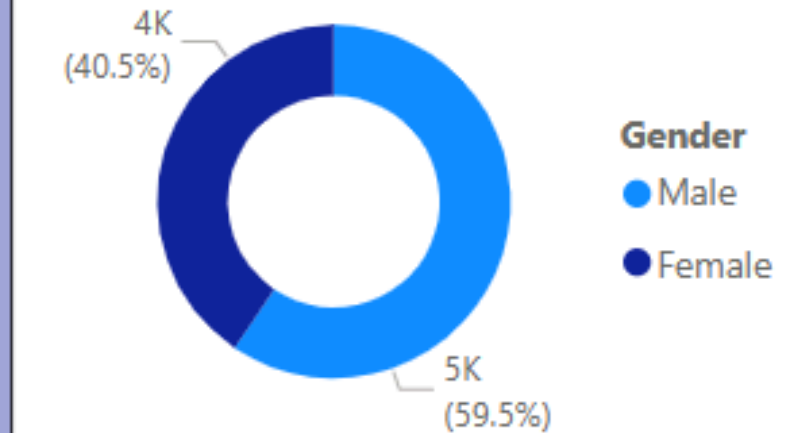
Smoking Impact Score by Gender



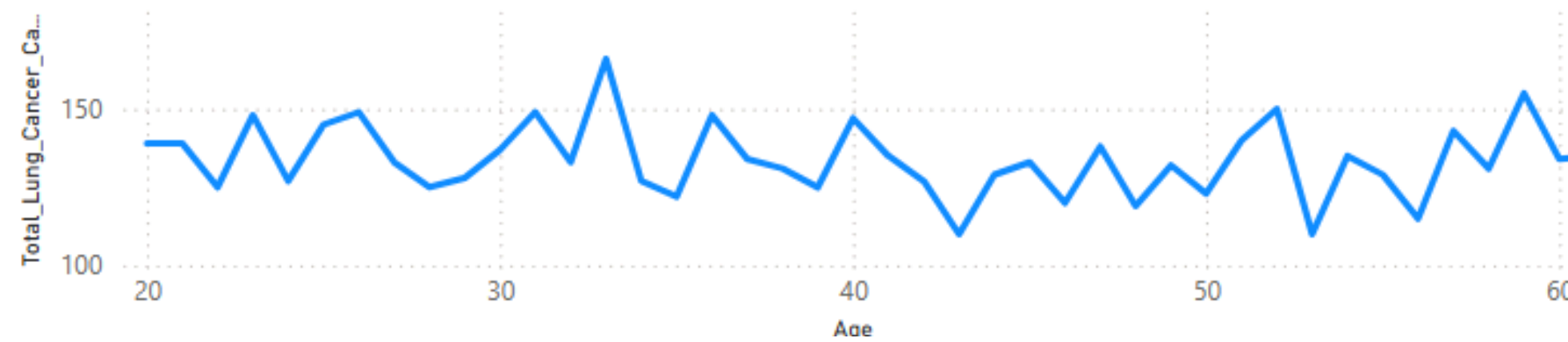
Mortality Rate vs. Air Pollution Exposure



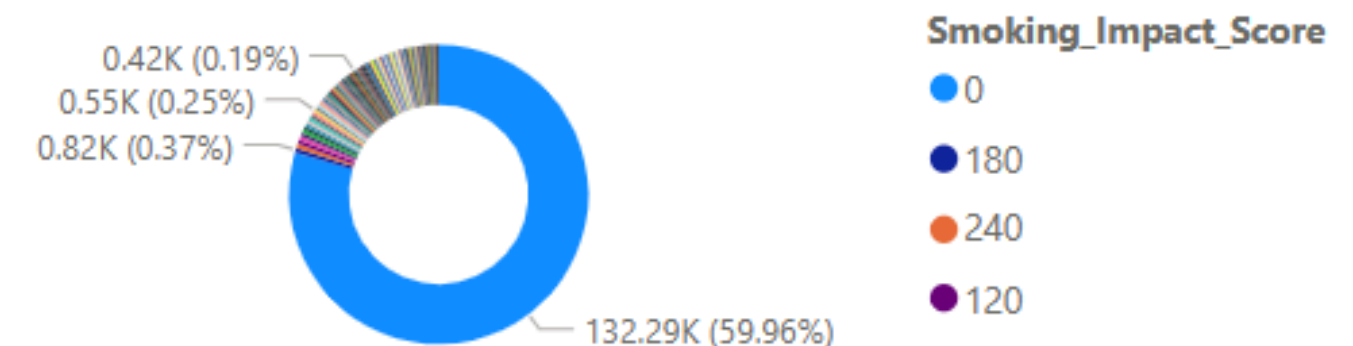
Lung Cancer Cases by Gender



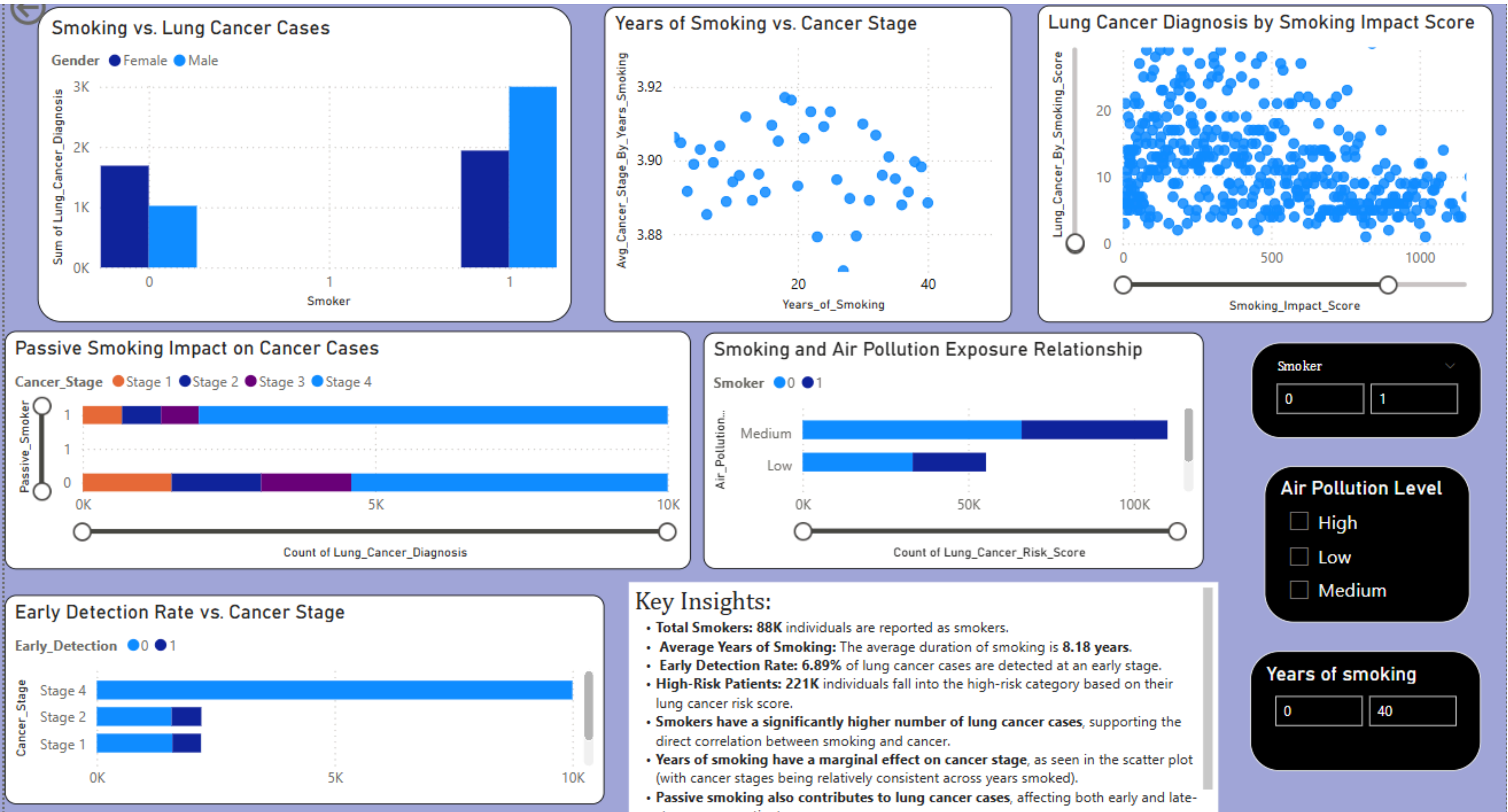
Age Distribution of Lung Cancer Patients



Lung Cancer Risk Score Distribution

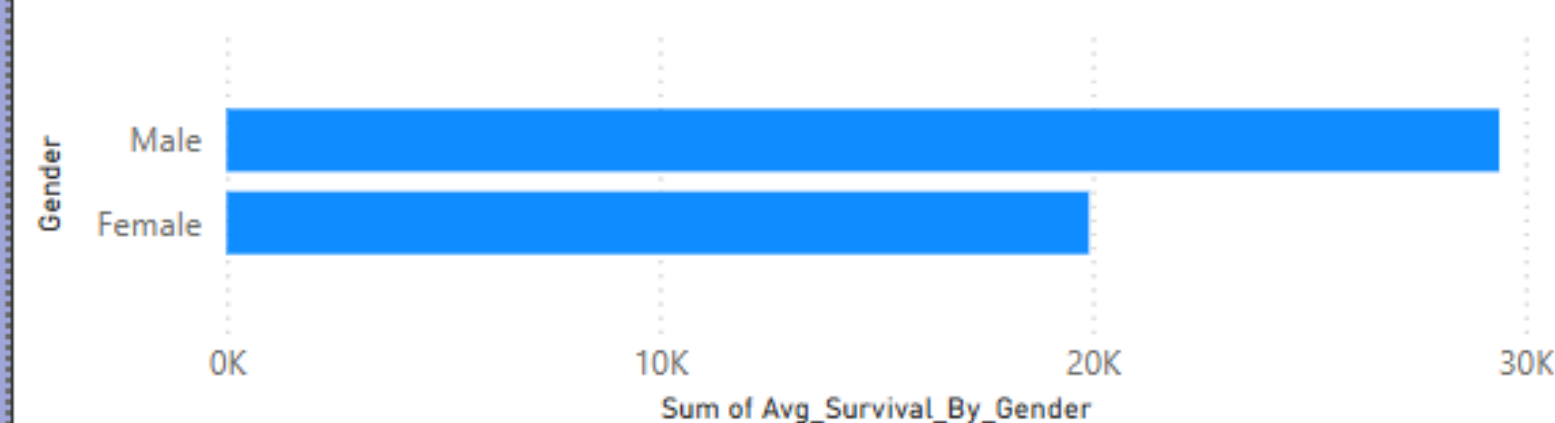


<Data Visualisation & Dashboard>

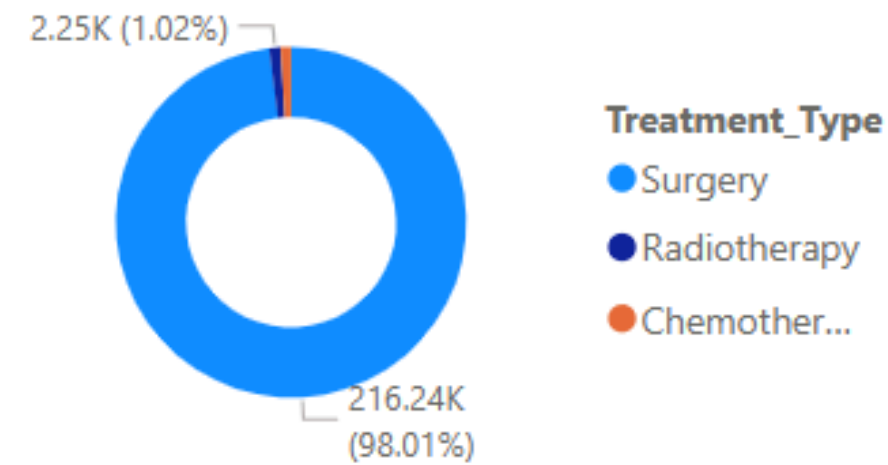


<Data Visualisation & Dashboard>

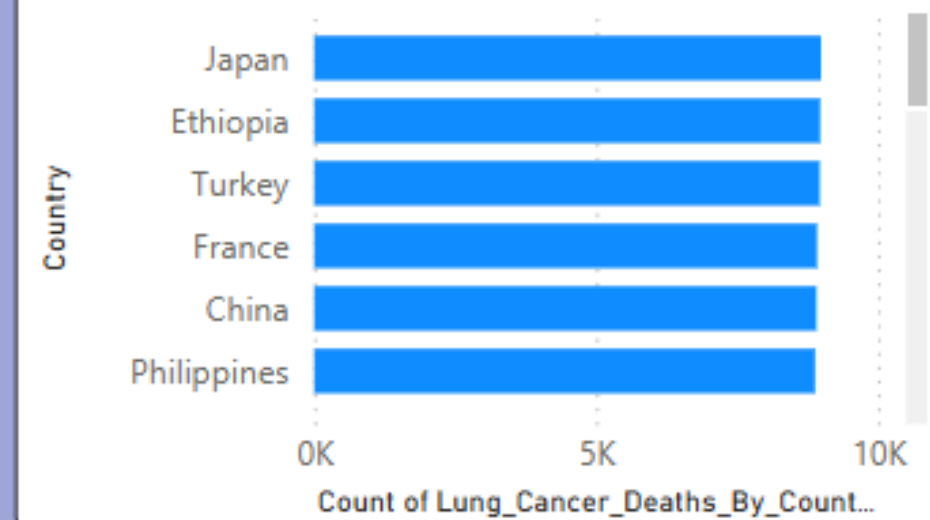
Survival years over gender.



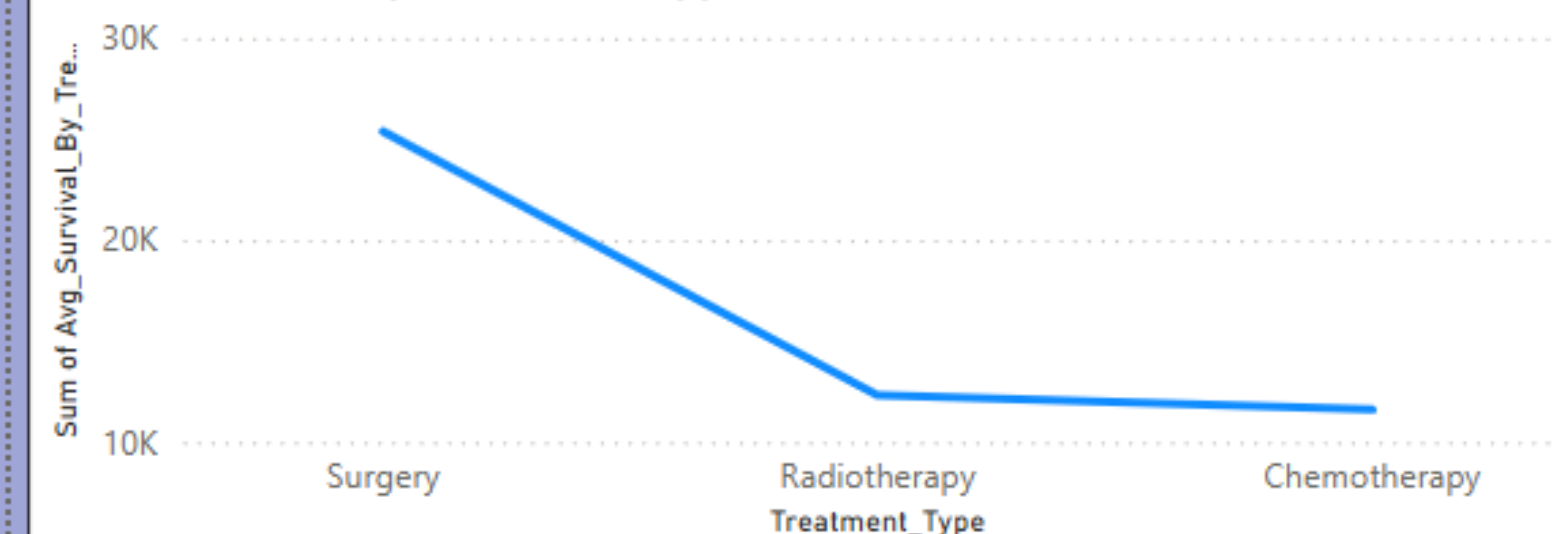
Mortality Rate vs. Treatment Type



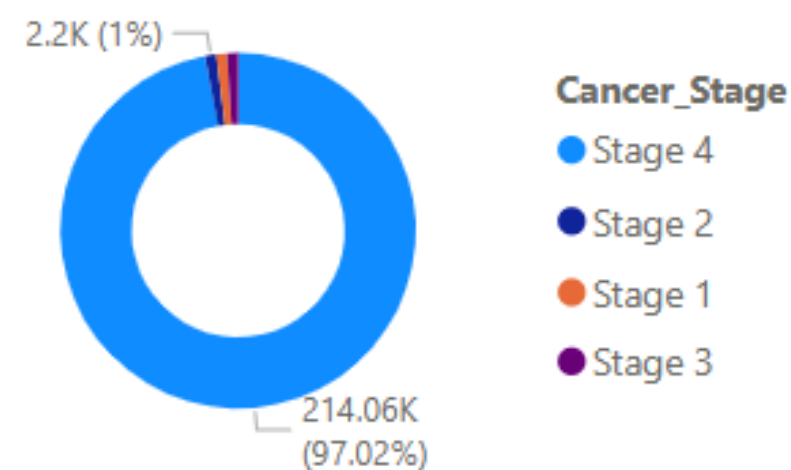
Lung Cancer Deaths by Country



Survival Years by Treatment Type



Survival Rate Distribution by Cancer Stage



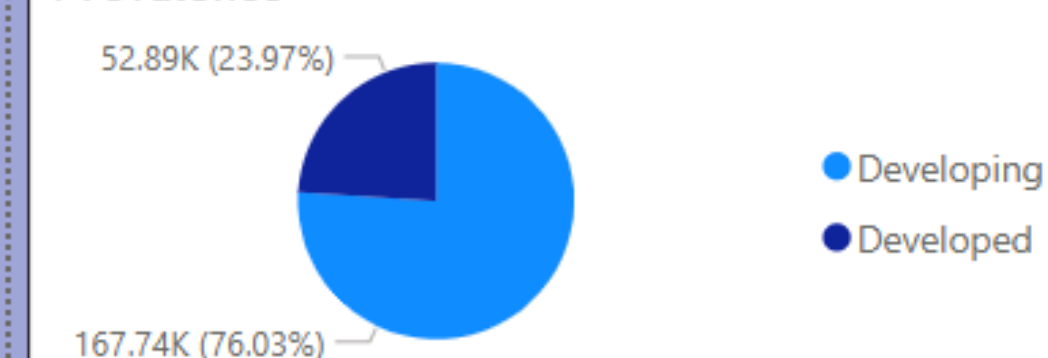
Developed or Developing

- ☐ Developed
- ☐ Developing

Treatment Type

- ☐ Chemotherapy
- ☐ Radiotherapy

Developed vs. Developing Countries: Cancer Prevalence



Key Insights:

- **Average Survival Years:** The average survival duration for lung cancer patients is **0.22 years**, indicating a very low survival rate post-diagnosis.
- **Total Lung Cancer Deaths:** The reported number of lung cancer-related deaths is **14 billion**, which seems unusually high and may indicate a data entry or scaling issue. ⚠️
- **Lung Cancer Prevalence:** The prevalence of lung cancer in the dataset is **1.50%**, highlighting its significant impact.
- **Survival Rate in Early Detection:** The survival rate among patients diagnosed early is **2.10%**, reinforcing the importance of early screening for better patient outcomes.
- **Surgery has the highest survival years**, whereas chemotherapy and radiotherapy show significantly lower survival durations.
- **The mortality rate is highest in developing countries**, suggesting a lack of access to advanced treatments.
- **Males have a slightly higher survival duration than females**, possibly due to differences in diagnosis and

Gender

- ☐ Female
- ☐ Male

Cancer Stage

- ☐ Stage 1
- ☐ Stage 2
- ☐ Stage 3

<Conclusion>

- **Smoking & Lung Cancer** – High correlation between smoking (active & passive) and lung cancer cases.
- **Environmental Impact** – Air pollution and occupational exposure significantly contribute to lung cancer risk.
- **Demographics** – Higher prevalence in individuals aged 60+ and a slightly higher occurrence in males.
- **Early Detection** – Leads to lower mortality rates and better survival outcomes.
- **Country & Economy** – Developed vs. developing countries show different prevalence trends; smoking rates impact cases.
- **Treatment & Survival** – Different treatments yield varying survival rates, with advanced-stage cases having lower survival.

Key Takeaways

- Early detection & awareness campaigns are crucial.
- Pollution control & smoking cessation policies can reduce cases.
- Targeted healthcare strategies improve patient outcomes.

< Business Recommendations >

Lung cancer poses a significant health burden, with smoking and environmental exposure as major risk factors. To improve patient outcomes and reduce mortality rates, businesses in healthcare, pharmaceuticals, and public health should focus on the following strategic initiatives:

- **Prioritize Early Detection Initiatives:**

Invest in advanced diagnostic technologies, such as AI-driven tools and low-dose CT scans, to enhance early lung cancer detection, particularly in high-risk populations.

- **Expand Smoking Cessation Programs :**

Launch awareness campaigns and provide support for smoking cessation through nicotine replacement therapies and digital interventions to reduce smoking-related lung cancer cases.

- **Leverage Data for Personalized Treatment :**

Invest in AI and predictive analytics to develop personalized treatment plans and support precision medicine approaches for better outcomes in chemotherapy and immunotherapy.

- **Promote Environmental Health & Access to Treatment :**

Collaborate with policymakers to reduce air pollution and enhance access to surgical treatments, ensuring affordable care in underserved regions